

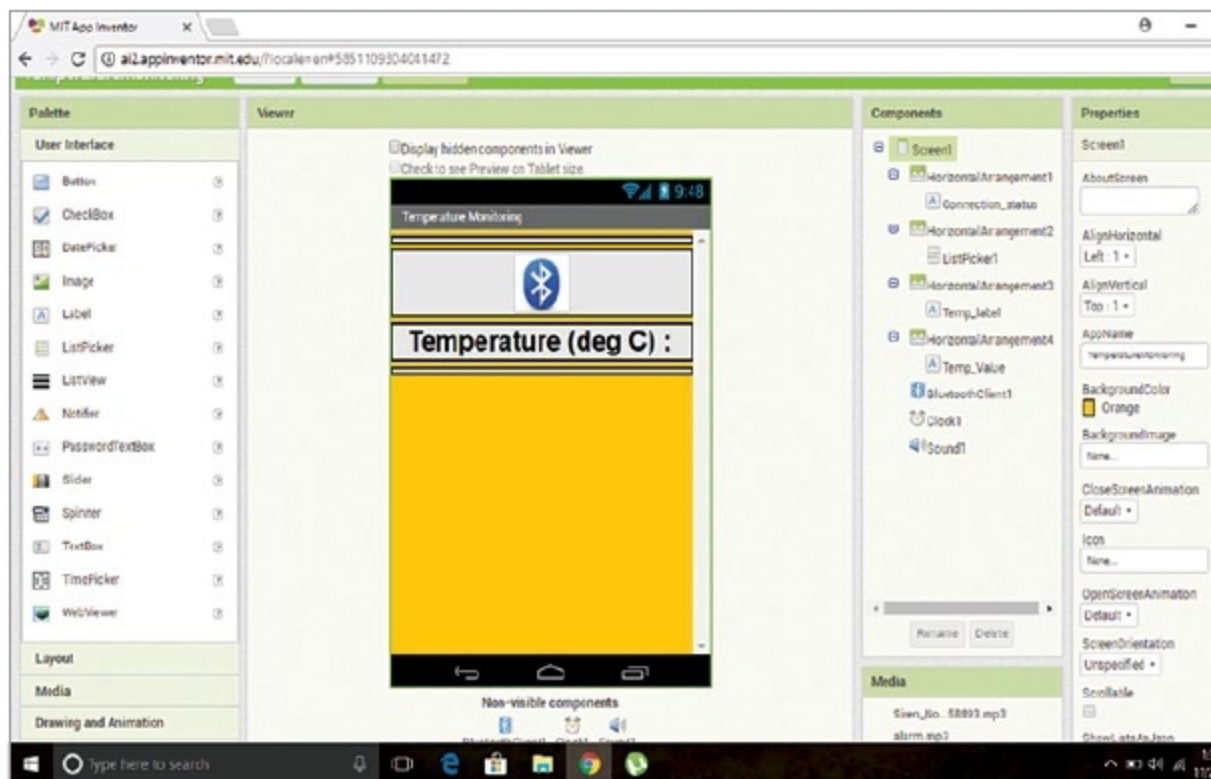


Fig. 4: Screenshot of Designer section as designed on MIT App Inventor

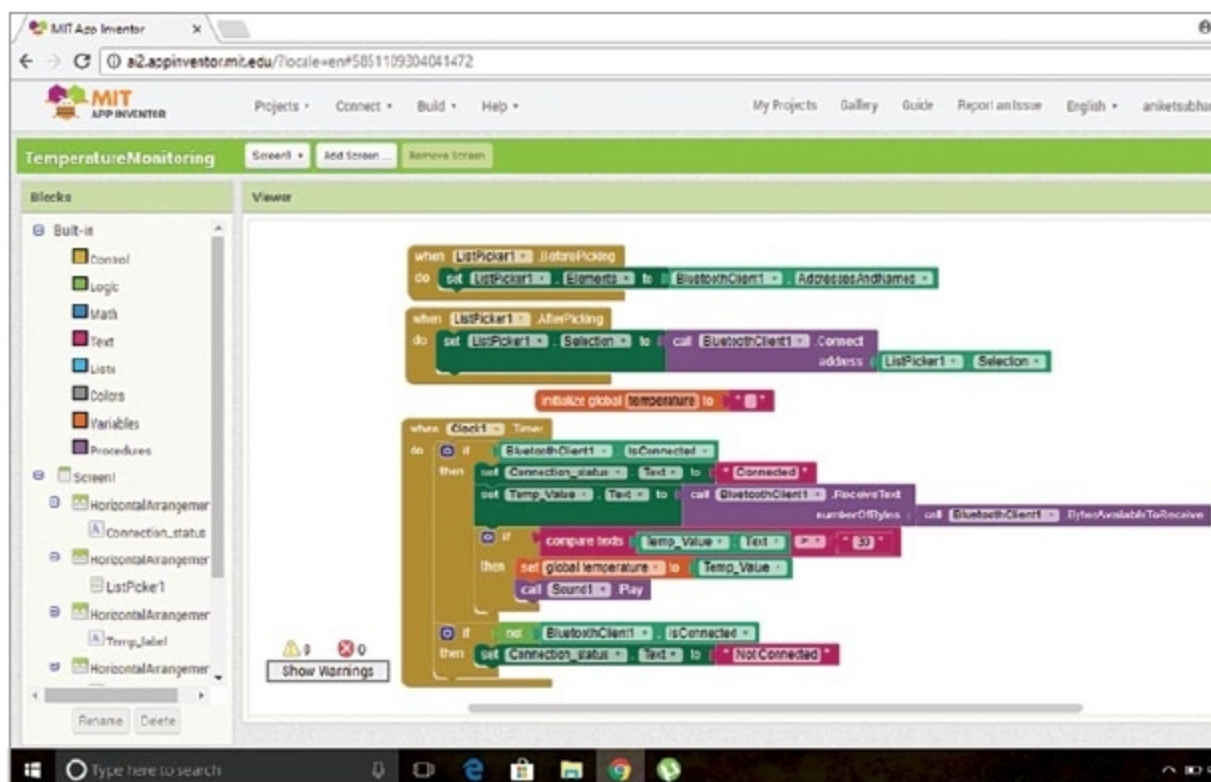


Fig. 5: Screenshot of Blocks section as designed on MIT App Inventor

(in °C), and transmit temperature values to Bluetooth module via Tx pin of Arduino.

HC-05 Bluetooth module. HC-05 module is an easy-to-use Bluetooth SPP (Serial Port Protocol) module designed for wireless serial connection setup. It is a fully-qualified Bluetooth V2.0 + EDR (Enhanced Data Rate) 3Mbps modulation with complete 2.4GHz radio transceiver and baseband. It uses CSR's BlueCore4-Ext single-chip Bluetooth system with CMOS technology and adaptive frequency hopping (AFH) feature.

In this project, Bluetooth module is used to wirelessly transmit sensed temperature to the smartphone.

Android smartphone. The over-temperature alert generation app is based on Android and has been designed using MIT App Inventor. MIT App Inventor is an intuitive, visual programming environment that allows the user to build fully-functional apps for smartphones and tablet PCs.

EFY Note

The source code of this project is available for free download at source.efymag.com

When this custom-designed app runs on the smartphone, data sent by HC-05 Bluetooth module is received and displayed on the smartphone. When sensed temperature exceeds critical temperature, an alarm is sounded on the smartphone. This app has been tested on Redmi 3S smartphone.

Testing procedure

The steps for testing are as follows:

1. Upload Arduino source code (bt.ino) file to Arduino. In the program, Serial.println () function helps transmit temperature value from Tx pin of Arduino to Bluetooth.

2. Install Android app Temp_Monit.apk on the smartphone. After successful installation, turn on Bluetooth connection of the smartphone. After successfully pairing HC-05 Bluetooth module with Bluetooth of the smartphone, open the app in the smartphone, and press the pushbutton with Bluetooth logo.

From the paired devices list, choose HC-05. After successful connection is established, 'Connected' will be displayed on the app. The Android application will display temperature data. A screenshot of temperature as observed on the smartphone is shown in Fig. 3.

Screenshots of Designer and Blocks sections, as designed using MIT App Inventor, are shown in Figs 4 and 5, respectively. If temperature exceeds critical value (here, 30°C is used for testing purposes), a buzzing sound will be produced on the smartphone.

While the Android source code uses a temperature of 30°C as over-temperature value for alert generation, you can use any other value of maximum temperature by changing number 30 in 'If compare texts' line of the code in Blocks section. **EFY**



Aniket Subham has a keen interest in MCU-based embedded systems



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